SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 1.3: Quadratic Equations — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.3: Quadratic Equations
Driving Question	DRIVING QUESTION: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems — like building the Harambee hall in Githurai?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring the Phenomenon: The Githurai Harambee Hall Problem**	Teacher presents the anchoring phenomenon: the Githurai community wants to build a rectangular Harambee hall with an area of 120 m². The width is w metres and the length is (w + 2) metres. Students observe that multiplying these two expressions — each containing w — produces the equation $w(w + 2) = 120$, which expands to $w^2 + 2w - 120 = 0$. The teacher displays this equation prominently and asks students to attempt to solve it using linear strategies they already know. Students observe that none of their familiar linear methods work.	Students learned that when two variable expressions are multiplied together and one of them contains the unknown, the result is an equation with a squared term (w^2). This makes the equation structurally different from linear equations of the form $ax + b = 0$. Specifically, the hall problem produces $w^2 + 2w - 120 = 0$ — a quadratic equation — which cannot be solved by simple inverse operations. Students learned that a new class of equation requires a new class of solving strategies, motivating the entire unit. Pedagogical note: Reinforce that w^2 is not the same as $2w$; use a quick area diagram on the board to show why $w \times w = w^2$ physically represents a square region.	The teacher connects the observed equation back to the real context: the hall dimensions are unknown, the area constraint is real, and the community cannot proceed with construction without knowing w. This frames the driving question for the unit. The teacher explains that the equation $w^2 + 2w - 120 = 0$ belongs to the family of quadratic equations (highest power of the variable is 2), and that over the coming lessons students will build a toolkit of methods to crack it. Teacher posts the equation on the Driving Question Board (DQB) and labels it 'Our Target Equation.' Pedagogical note: Cold-call at least three students to articulate in their own words why the equation has a w^2 term before moving on — this checks for the core conceptual shift of the lesson.
Lesson 2 **Expanding and Identifying Quadratic Expressions**	Teacher guides students through expanding products of two linear brackets, e.g. $(x + 3)(x + 5)$, $(2x - 1)(x + 4)$, and $(w + 12)(w - 10)$, using the distributive property (FOIL). Students observe each step on the board and record results. Critically, the teacher	Students learned that: (a) expanding two linear brackets always produces three types of terms — an x^2 term, an x term, and a constant term — which together form a quadratic expression in standard form $ax^2 + bx + c$; (b) the coefficients a, b, and c come	The teacher explains that expansion is a one-way journey from factored form to standard form, and poses the reverse question: if we already have $w^2 + 2w - 120 = 0$, can we work backwards to find the brackets? This reverse process —

	expands $(w + 12)(w - 10)$ and students observe it produces $w^2 + 2w - 120$ — exactly the Githurai hall equation. Students also observe examples of expressions that are NOT quadratic (linear, cubic) to sharpen their ability to identify the defining feature: a squared variable term with no higher power.	directly and predictably from the numbers in the original brackets; (c) the Githurai hall equation $w^2 + 2w - 120 = 0$ is equivalent to $(w + 12)(w - 10) = 0$, a fact that will be exploited in Lesson 3. Pedagogical note: Have students verify the expansion of $(w + 12)(w - 10)$ independently before revealing the connection to the hall problem — the 'aha moment' is more powerful when students discover it themselves.	factorisation — is the subject of Lesson 3. The teacher updates the DQB: 'We now know the hall equation can be written as a product of two brackets. Next question: how do we find those brackets systematically?' Pedagogical note: Emphasise that $a = 1$ (the coefficient of w^2) in the hall problem, which makes factorisation straightforward — students will encounter $a \neq 1$ cases later.
Lesson 3 **Factorisation of Quadratic Expressions**	Teacher presents several quadratic expressions (e.g. $x^2 + 7x + 12$, $x^2 - 5x + 6$, $x^2 + 2x - 120$) and leads students through the factor-pair search: identify two integers that multiply to c and add to b. Students work in pairs to complete a factor-pair table for each expression, then rewrite each as a product of two brackets. The teacher deliberately includes the hall expression $x^2 + 2x - 120$ so students factorise it themselves and confirm $(x + 12)(x - 10)$.	Learners learned that: (a) factorisation is the reverse of expansion; (b) to factorise $x^2 + bx + c$, find two integers p and q such that $p \times q = c$ and $p + q = b$, then write $(x + p)(x + q)$; (c) the sign of c determines whether p and q have the same sign ($c > 0$) or opposite signs ($c < 0$), which is a reliable checking heuristic; (d) not every quadratic expression factorises neatly over the integers — some require other methods. Pedagogical note: The factor-pair table is a powerful scaffold; encourage students to keep using it rather than guessing. For the hall problem, $c = -120$, so students must find a positive and a negative factor — connecting sign rules to real context reinforces meaning.	The teacher explains that factorisation transforms a quadratic expression into a product form, which is the gateway to solving quadratic equations (Lesson 4). The teacher updates the DQB: 'We can now factorise the hall expression. But how does factorisation actually give us the value of w?' Pedagogical note: Assess whether students can independently factorise at least two expressions before ending the lesson. A quick exit ticket (factorise $x^2 - 3x - 28$) takes 3 minutes and reveals readiness for Lesson 4.
Lesson 4 **Solving Quadratic Equations by Factorisation**	Teacher returns to the Githurai hall equation: $w^2 + 2w - 120 = 0$. Students have already factorised the left side as $(w + 12)(w - 10)$. The teacher introduces the Zero Product Property: if $A \times B = 0$, then $A = 0$ OR $B = 0$. Students observe the teacher apply this to set $w + 12 = 0$ and $w - 10 = 0$, yielding $w = -12$ and $w = 10$. Students then observe that $w = -12$ is rejected because a dimension cannot be negative in the real context, leaving $w = 10$ m as the hall width and $(10 + 2) = 12$ m as the length. Students verify: $10 \times 12 = 120$ ✓.	Students learned that: (a) once a quadratic equation is factorised, the Zero Product Property converts it into two simple linear equations; (b) each linear equation gives one potential solution (root) of the quadratic; (c) a quadratic equation has at most two roots; (d) in real-world problems, solutions must be checked against context constraints — negative lengths are mathematically valid but physically meaningless. Pedagogical note: The moment of solving the driving question's core equation is emotionally significant — pause, celebrate it, and have a student read aloud the final answer in a full sentence ('The Githurai hall is 10 metres wide and 12 metres long'). This anchors the purpose of the entire method.	The teacher explains that factorisation + Zero Product Property is the most elegant solving method when it works, but it requires the quadratic to factorise over the integers. The teacher poses a challenge: 'What if the area had been 115 m²? Would $w^2 + 2w - 115 = 0$ factorise neatly?' Students attempt to find factor pairs and discover it does not — motivating the need for alternative methods introduced in Lessons 5 and 6. The DQB is updated: 'Solved! But only when factorisation is possible. What do we do when it isn't?'

Lesson 5 **Solving Quadratic Equations by Completing the Square**	Teacher presents a quadratic equation that does not factorise over integers (e.g. $x^2 + 6x - 7 = 0$, then a context example such as a Kericho tea-farm plot with area that gives $x^2 + 4x = 12$). Students observe the teacher construct a geometric square model: $x^2 + 4x$ is represented as a square of side x plus a rectangle of width 4, and the 'missing corner' needed to complete the square is $(4/2)^2 = 4$. Students observe the teacher add 4 to both sides, rewrite as $(x + 2)^2 = 16$, then take the square root of both sides to find $x + 2 = \pm 4$, giving $x = 2$ or $x = -6$.	Learners learned that: (a) any quadratic equation of the form $x^2 + bx = c$ can be rewritten as $(x + b/2)^2 = c + (b/2)^2$ — a process called completing the square — by adding $(b/2)^2$ to both sides; (b) this converts the equation into a perfect square on the left, allowing a direct square root; (c) taking the square root introduces a ± sign, yielding two solutions; (d) the method works for any quadratic, not just those with integer factor pairs. Pedagogical note: The geometric square diagram is essential — it gives students a visual anchor for why $(b/2)^2$ is added, not an arbitrary trick. Draw it large on the board and leave it visible throughout the lesson.	The teacher explains that completing the square is both a solving technique and the foundation for deriving the quadratic formula in Lesson 6. Students revisit the Githurai hall equation ($w^2 + 2w - 120 = 0$) and solve it by completing the square as a verification exercise, confirming they get $w = 10$ again. The DQB is updated: 'Completing the square always works — but is there an even faster general method?' Pedagogical note: Ensure students can execute the 'add $(b/2)^2$ to both sides' step fluently before Lesson 6; a common error is forgetting to add to the right side as well.
Lesson 6 **The Quadratic Formula & The Mombasa Stone Problem**	Teacher derives the quadratic formula live in front of students by completing the square on the general equation $ax^2 + bx + c = 0$. Students observe each algebraic step annotated on the board, arriving at $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$. Teacher then presents the Mombasa stone-throwing context: a stone is thrown upward from the Fort Jesus cliff with height modelled by $h = -5t^2 + 20t + 25$. Students observe the teacher substitute $h = 0$ (stone hits the ground), rearrange to $5t^2 - 20t - 25 = 0$, identify $a = 5$, $b = -20$, $c = -25$, and substitute into the formula, obtaining $t = 5$ seconds (rejecting the negative root).	Students learned that: (1) the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$ can be derived by completing the square on any quadratic equation in standard form, which means it works universally — for all quadratics, factorisable or not; (2) the expression under the square root, $b^2 - 4ac$ (the discriminant), controls how many real solutions exist: positive → two solutions, zero → one solution, negative → no real solutions; (3) the formula is a direct computational tool — once a, b, and c are correctly identified, solutions are mechanical to find. Pedagogical note: Make students write 'a = __, b = __, c = __' explicitly before substituting — this single habit prevents the majority of formula errors. Cold-call three students to state a, b, c for different equations before they calculate.	The teacher explains that the quadratic formula is the ultimate fallback: every quadratic can be solved with it. However, factorisation (when it works) is faster, and completing the square builds deeper understanding. Students are guided to choose methods strategically: try factorisation first; if factor pairs are not obvious within 60 seconds, use the formula. The DQB is updated: 'We now have three methods. When do we use each one?' Students add a method-selection guide to their notes. The Githurai hall problem is solved again using the formula as a final verification, cementing the connection to the driving question.
Lesson 7 **Applying Quadratic Equations to Community Planning Problems**	Teacher presents a set of real Kenyan community planning scenarios — e.g. a rectangular maize farm in the Rift Valley whose area and perimeter relationships yield a quadratic, a water tank design in Kisumu, and a githeri-serving table layout for a school in Nairobi. Students work in groups to model each scenario as a quadratic equation, choose an appropriate solving method, solve, reject non-physical	Students learned to apply the full quadratic equation toolkit — factorisation, completing the square, and the quadratic formula — to authentic community planning contexts. They learned that the modelling cycle has four stages: (1) define the unknown and express all given information algebraically; (2) form the quadratic equation from the area, perimeter, or other relationship; (3) solve using the most efficient method; (4)	The teacher brings the unit to a close by returning to the Githurai Harambee hall problem. A student volunteer solves it on the board from scratch — choosing their preferred method — and presents the final answer to the class as if addressing the community committee. The teacher facilitates a DQB review: the class revisits every question posted since Lesson 1 and confirms each has been answered.

	solutions, and present their findings as a community recommendation (dimensions in metres, quantities in units).	interpret and validate solutions against real-world constraints. Students also consolidated method-selection strategy: factorise when integer factor pairs are obvious; complete the square when the leading coefficient is 1 and b is even; use the formula otherwise. Pedagogical note: Emphasise stage 4 repeatedly — a negative width or a time before the event starts are mathematical solutions that must be rejected with a written reason.	Students write a one-paragraph 'Community Report' summarising the hall dimensions and explaining why the quadratic equation was necessary. Pedagogical note: The Community Report serves as a formative assessment of both mathematical fluency and communication competency. Look for correct dimensions ($w = 10$ m, $l = 12$ m), correct rejection of $w = -12$, and a clear sentence explaining what a quadratic equation is and why a linear approach would have failed.